Monitoring the Quality of Internet of Things Services

Abstract

The Internet of Things (IoT) refers to a network of physical objects embedded with sensors, software, and other technologies that connect and exchange data with other devices and systems over the internet. The growing complexity of IoT calls for reliable methods to monitor the quality and performance of interconnected devices. This thesis investigates the use of the Matrix Profile, a simple yet powerful technique for time-series analysis, to detect anomalies that indicate service degradation in IoT systems. The proposed approach is evaluated on univariate and multivariate datasets and compared against established statistical and AI-based algorithms assess its effectiveness, interpretability, and computational efficiency. *[Note: This section will later include the main findings and comparative results once the evaluation is completed.]*

Keywords

Anomaly detection, Internet of Things, Matrix Profile, Time-Series Analysis

Introduction

The IoT connects millions of sensors and smart devices that communicate and share information without human intervention. These systems continuously produce large volumes of time-series data that need to be processed almost in real time. Analyzing such data efficiently requires methods that can work under limited computational resources while maintaining accuracy and reliability.

In IoT environments, two major challenges often arise: concept drift and anomaly detection. Concept drift occurs when the statistical behavior of data changes over time, making previously trained models less accurate [1]. Anomaly detection, also known as outlier or deviation detection, refers to identifying unusual patterns that deviate from the expected system behavior [2]. The ability to recognize unexpected behaviors has led to the development of numerous anomaly-detection algorithms in the literature, ranging from classical statistical approaches to modern machine-learning and artificial-intelligence-based techniques [3]. Recently, the Matrix Profile (MP) framework has emerged as a powerful, data-driven approach for time-series analysis. It was first introduced by Yeh et al. [4] and has since attracted considerable attention due to its versatility, simplicity, and scalability.

This thesis focuses on evaluating the MP as a lightweight and explainable method for detecting anomalies in IoT sensor data. The study applies MP to both univariate and multivariate IoT datasets and compares its performance with established statistical and machine-learning algorithms to assess its effectiveness for IoT quality monitoring.

The rest of the thesis is organized as follows. Section 2 reviews the background and related work on IoT anomaly detection. Section 3 describes the research methodology used in this study. Section 4 provides a detailed description of the datasets employed. Section 5 reports and analyzes the experimental evaluation results. Finally, Section 6 presents the conclusions and outlines possible directions for future work.

Background & Related Work

This section provides an overview of the fundamental concepts underlying anomaly detection in IoT. It begins by defining the key characteristics of time-series data, followed by an explanation of concept drift and the different types of anomalies. The section then reviews related research, summarizing major algorithms, methodologies, and recent developments in the field, with particular emphasis on approaches relevant to time-series and IoT data analysis.

Time-Series

Time-series data consist of sequences of observations collected at uniform time intervals, capturing how a given phenomenon evolves over time. This temporal structure allows the analysis of dynamic behaviors and helps identify how values change in chronological order. Each observation x_t represents the state of a system at time t, and examining the full sequence enables the discovery of patterns such as long-term trends, seasonality, cycles, and irregular fluctuations [5]. As noted in [6], time series are often divided into univariate (one-dimensional) and multivariate (multi-dimensional) which are described below.

Univariate Time Series

A univariate time series (UTS) is a sequence of data points corresponding to a single variable that changes over time. It can be expressed as:

$$X = (x_1, x_2, \ldots, x_t)$$

where x_i represents the data value at timestamp $i \in T$, and $T = \{1,2, \ldots, t\}$.

Multivariate Time Series

A multivariate time series (MTS) represents multiple variables that evolve together over time, where each variable is influenced by both its past values (temporal dependency) and by correlations with other variables (spatial or inter-metric dependency). An MTS can be represented as a sequence of vectors over time:

$$X = (X^1, X^2, \ldots, X_{\mathrm{t}}) = \{(x_1^1, x_1^2, \ldots, x_1^{\mathrm{d}}), (x_2^1, x_2^2, \ldots, x_2^{\mathrm{d}}), \ldots, ((x_{\mathrm{t}}^1, x_{\mathrm{t}}^2, \ldots, x_{\mathrm{t}}^{\mathrm{d}})\}$$

where $X_i = (x_{\mathrm{i}}^1, x_{\mathrm{i}}^2, \ldots, x_{\mathrm{i}}^{\mathrm{d}})$ represents a data vector at time i,

each x_i^j indicates the observation at time i for the j^{th} dimension,

and $j = 1,2, \ldots d$, where d is the total number of dimensions.

Concept drift

Concept drift is an evolution of data that invalidates an existing model when the underlying data distribution changes over time [7]. In the context of classification learning [8], let $P(Y)$ denote the prior probability distribution over the class labels, $P(X)$ the prior probability distribution over independent variables (covariates), $P(X, Y)$ the joint probability distribution over objects and class labels. The conditional probability $P(Y \mid X)$ represents the likelihood of observing a class label given the input features, whereas $P(X \mid Y)$ describes the posterior distribution of independent variables conditioned on class labels.

To capture temporal dynamics, $P_t(X)$ denotes the probability distribution at time t. Given a joint distribution (concept) $P(X, Y)$ where Y is the target variable (if applicable), concept drift occurs between times t and $u = t + \Delta$ when the distributions change:

$$P_t(X, Y) \neq P_u(\mathrm{X}, \Upsilon)$$

where Δ represents the difference in time (time interval) between the two observations t and u.

Concept drift can be categorized into several types [7]. Real drift occurs when the target concept itself changes, altering the relationship between inputs and outputs. Virtual drift involves changes in the distribution of input features without affecting the underlying concept. Class prior drift refers to variations in the proportion of class labels over time. When such changes occur suddenly, the phenomenon is known as abrupt drift; when they evolve gradually, it is called gradual drift. Finally, recurrent drift describes cases in which previously observed concepts reappear periodically, often due to seasonal or cyclic patterns.

Anomalies

Anomalies are occurrences in a dataset that are unusual and do not conform to a well-defined notion of normal behavior [9] [10]. In the IoT domain, they refer to deviations from expected behavior within devices, networks, or sensor data streams. Detecting such anomalies is crucial for maintaining the integrity, reliability, and security of IoT services. As the number of connected devices increases, identifying abnormal patterns quickly and accurately becomes essential for ensuring service quality and preventing potential faults or attacks. Anomalies are commonly categorized into three main types: global (or point), contextual, and collective [11].

Global or point anomalies refer to individual data points that significantly deviate from the rest of the dataset. In IoT networks and wireless sensor systems, such outliers can generally be classified as errors, events, or malicious attacks [12]. Errors are typically caused by faulty sensor readings or random noise and, although common, they reduce the reliability of collected data. Events, on the other hand, represent meaningful environmental changes such as temperature spikes, air quality variations, or natural phenomena, which are often sustained over time. Malicious attacks occur when compromised sensors intentionally transmit false data, posing serious security risks to the entire network. Contextual anomalies occur when the anomaly is not defined solely based on the individual data point but depends on the context or the relationship between data points. Collective anomalies involve anomalous patterns or behaviors that can only be identified by considering a group or collection of data points together. In multivariate time series data, relationships among sensors and variables are complex [13], making it less precise to classify anomalies as point, contextual, or collective. Even if individual sensor readings appear normal, a combined analysis might reveal patterns inconsistent with typical inter-sensor dependencies.

Related work

Once the importance of anomaly detection in IoT services has been established, the next critical step is to examine the existing landscape of techniques and algorithms used for this purpose. Researchers have proposed a wide variety of methods ranging from formal-logic monitoring to machine-learning and deep-learning approaches.

A recent comprehensive survey by the authors of [14] emphasizes the importance of anomaly detection for ensuring the safety and reliability of smart environments. Their study provides an extensive overview of state-of-the-art anomaly detection techniques applied to multivariate time-series data in domains such as smart homes, smart transport, and smart industry. The authors review a wide range of methods—spanning statistical, proximity-based, and deep-learning approaches—and also discuss key aspects such as anomaly categorization, detection scenarios, evaluation metrics, and available benchmark datasets. This work highlights the increasing research focus on adaptive and explainable methods that can effectively handle large-scale, dynamic IoT data streams.

Beyond these general frameworks, several research efforts have explored specific algorithms and models for monitoring and anomaly detection in IoT systems. In [15], the authors present an offline monitoring algorithm for Spatio-Temporal Reach and Escape Logic (STREL) and demonstrate its application in monitoring a simulated mobile ad-hoc sensor network. Their future work focuses on distributed monitoring algorithms to extend STREL for large-scale cyber-physical systems and on studying its expressive power for spatio-temporal reasoning. In [16], a framework is introduced for learning Signal Temporal Logic (STL) specifications from labeled datasets of normal and anomalous trajectories. It uses a robust genetic algorithm (ROGE) that combines an Evolutionary Algorithm (EA) for formula-structure learning with a Gaussian-process-based optimization for parameter synthesis. The algorithm appears to be specialized for certain types of anomaly detection and classification tasks.

Machine learning has been widely used to detect outliers and abnormal patterns in IoT data. The study in [17] tackles the critical challenge of outlier detection within the context of Internet of Things (IoT) frameworks and Wireless Sensor Networks (WSNs), emphasizing the unreliability of sensor data and the potential of machine-learning techniques to overcome it. Within the realm of machine learning-based outlier detection, two primary categories of techniques come to the forefront: statistical-based methods and classification-based methods. The future trajectory of this work involves employing ensemble methods for outlier detection in IoT and WSN, utilizing machine learning techniques and semantics to distinguish between errors and events, and integrating offline and online learning techniques to enhance outlier detection. The paper in [18] proposes an online anomaly detection algorithm called OnlineBPCA, based on Bilateral Principal Component Analysis (B-PCA), to address the challenge of real-time traffic anomaly detection in network management. They introduce several novel techniques within OnlineBPCA to support quick and accurate anomaly detection. The authors also express their intention to extend these algorithms to detect data anomalies in various domains, including mobile and wireless sensor networks, wireless communications, social networks, recommendation systems, and other Internet of Things (IoT) systems in their future work. In [19], the authors leverage deep learning methods to design an effective intrusion detection model for IoT networks. They employ convolutional neural network (CNN) architectures for multiclass and binary traffic classification, enabling the detection of various attack types within IoT network traffic.

The work in [20] highlights the significance of Multivariate Time Series Anomaly Detection (MTSAD) in the context of the growing deployment of sensors in various environments. It reviews the current state of the art in unsupervised MTSAD, including advanced machine learning and signal processing techniques, such as deep learning. The joint analysis of temporal patterns and measurement dependencies across different sensors (often exhibiting complex behavior) makes MTSAD a challenging problem. However, there are several challenges and promising directions for further research in unsupervised Multivariate Time Series Anomaly Detection, as many methods are specific and tailored to particular use cases and no one-size-fits-all approach is available. In [21], stream-based machine learning models are applied for the detection of various types of network attacks using real network traffic measurements from the WIDE backbone network, a high-capacity, central infrastructure that connects multiple smaller networks, such as local area networks (LANs) or wide area networks (WANs), across large geographical areas. The paper emphasizes the need for real-time analysis of network monitoring data and the findings suggest that certain machine learning models can effectively handle concept drift in real-time network data streams, contributing to improved network security and anomaly detection.

The study in [22] further emphasizes that the continuous nature of IoT data and the need for online prediction, adaptation to concept drift, and early detection make traditional anomaly-detection methods unsuitable for real-time use. They advocate for continuous-learning algorithms that minimize false positives and negatives.

Large-scale benchmarking is presented in [23], where 71 state-of-the-art anomaly detection algorithms were evaluated across diverse domains using 976 time series datasets. The findings indicate that, despite their higher training costs, deep-learning models are currently not competitive in anomaly detection, performing comparably to or worse than simpler statistical approaches.

Complementary to these algorithmic approaches, [3] provides a detailed taxonomy of existing anomaly-detection methods, categorizing them according to their analytical principles. Statistical approaches typically distinguish normal from abnormal data by defining confidence boundaries, flagging values outside these limits as anomalies. While these methods are simple and interpretable, they often fail to detect subtle or weak deviations. Clustering-based algorithms, such as k-means or DBSCAN, assume that normal data points form clusters, while points that do not belong to any cluster are considered anomalous. However, when abnormal data form their own cluster, these methods may incorrectly classify normal points as noise. Density-based techniques, including Outlier Detection using In-degree Number (ODIN), Relative Density-based Outlier Score (RDOS), and Local Outlier Factor (LOF), identify anomalies by comparing the local density of each data point with that of its neighbors. Despite their accuracy, their computational cost makes them less suitable for real-time engineering systems. Ensemble methods, such as Isolation Forest, are efficient and widely used due to their linear time complexity. However, they mainly detect point anomalies and rely heavily on feature selection, which can make the results unstable. In contrast, distance-based approaches are easier to implement and do not require explicit feature extraction, making them well suited for time-series and signal-monitoring applications.

Within the category of distance-based approaches, one method in particular has gained attention for its versatility and interpretability—the Matrix Profile (MP). The MP is a powerful data structure designed to measure similarity between subsequences in time-series data, making it highly effective for detecting anomalies in both univariate and multivariate datasets [4]. It enables the efficient discovery of motifs—repeated patterns—and discords—unusual subsequences—within large data streams. Several studies have extended the Matrix Profile framework to enhance its performance and broaden its applications. For example, “Matrix Profile I” [4] introduced the concept through all-pairs similarity joins for time series, while “Matrix Profile II” [24] improved computational efficiency using GPU acceleration. “Matrix Profile V” [25] incorporated domain knowledge into the matching process, and “Matrix Profile VII” [26] explored time-series chains to capture evolving patterns over time. “Matrix Profile XX” [27] expanded its capability to visualize motifs of varying lengths, and “Matrix Profile XXVIII” [28] extended the method to handle multidimensional data through a K-of-N detection strategy. Building on these advances, “Matrix Profile XXIV” [29] proposed DAMP, an innovative algorithm that enhances discord computation and further demonstrates MP’s utility in time-series anomaly detection. A comprehensive overview of these developments is available in [30].

Further extending this line of research, in [31], the Interval Matrix Profile (IMP) is proposed as an enhanced version of MP designed to analyze periodic and seasonal time series. Their work addresses a key limitation of the classical MP, which does not explicitly account for seasonality and may therefore miss recurrent seasonal anomalies. The IMP performs flexible interval-based comparisons across different seasonal cycles, enabling the detection of anomalies that occur periodically—an important feature for climate and environmental datasets. The authors also introduced a constrained k-Nearest Neighbor variant to identify anomalies that appear across multiple time periods, such as recurring extreme weather events. In addition, they developed a scalable, block-based implementation that leverages caching, vectorization, and parallelism to improve performance. Tested on climate temperature datasets, their method effectively captured seasonal anomalies and pattern disappearance, outperforming the standard MP in both accuracy and computational efficiency.

In summary, the literature on anomaly detection in IoT and time-series analysis demonstrates rapid methodological evolution—from rule-based and statistical models to data-driven and explainable frameworks such as the Matrix Profile. Recent advancements, including the Interval Matrix Profile, highlight a shift toward more adaptive and seasonality-aware techniques, which are particularly relevant for real-world IoT scenarios where data streams are dynamic, multidimensional, and often periodic.

Methodology

Description of Datasets

Experimental Evaluation

Conclusions & Future Work

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